

MCA

PYTHON PRACTICE QUESTIONS

Q-1 Write a Python program that accepts a string from the user and performs the following operations:

- Display the string in reverse order.
- Count the number of vowels and consonants in the string.
- Check whether the string is a palindrome.
- Replace all vowels in the string with * and display the modified string.

Q-2 Write a Python program to develop a Hotel Reservation System with the following requirements:

- Store details of N rooms using a list of dictionaries, where each room has: room number, room type, price per day, and availability status
- Display details of all available rooms
- Display room number and price as a tuple for rooms of a given room type
- Identify and display the room(s) having the maximum price per day.
- Allow booking of a room using room number and update the availability status

Q-3 Write a python program to check whether the given number is prime or not? If prime then print the multiplication table of it.

Q-4 Write a Python program that:

- Displays all the words present in the sentence
- Displays only the **unique words**
- Displays the **count of unique words**

Q-5 Write a Python program for a **Library Management System** with the following requirements:

- Accept details of **N books**, including book ID, book name, author name, category, and number of copies available
 - Display all books belonging to a given category
 - Identify and display the book(s) having the **maximum available copies**
 - Display names of authors whose books have **less than a given number of copies**
 - Allow issuing a book and update the availability accordingly
 - Display the updated book details

Q-6 Write a Python program that accepts a number from the user and checks whether it is a **palindrome**. If it is a palindrome, display numbers from **1 to that number**; otherwise, display an appropriate message.

Q-7 Write a program in python to enter the string. Perform the following tasks

- 1) Count the number of times each word is repeated in the string
- 2) Count the number of palindrome words

Q-8 Write a Python program to develop an **Employee Management System** with the following requirements:

- 1) Store details of N employees using a suitable data structure (employee ID, name, department, basic salary)
- 2) Calculate and display the gross salary of each employee
- 3) Display details of employees having the highest gross salary
- 4) Allow updating the salary of an employee using employee ID
- 5) Count the number of employees in each department.

Q-9 Write a program to input n 3 digit numbers from the user. Find the sum of the digits of each number and store it in dictionary,